

Alex's Quest

Overview

Alex's Quest is a story-driven English learning mobile application designed to help Grade-school learners master sentence construction, grammar, and pronunciation by guiding a parrot named Alex through five chapters of an adventurous city-to-jungle quest. Built with modern cross-platform development practices using Expo SDK 55, React Native 0.83, and on-device Whisper speech recognition for real-time pronunciation scoring.

Features

Learning Modules

- **5 Story-Based Chapters**, each tied to a Learning Objective
 - Chapter 1: The Bakery District — *LO1: Sentence Construction*
 - Chapter 2: Fountain Square — *LO2: Subject–Verb Agreement*
 - Chapter 3: The Farmlands of Plenty — *LO3: Verb Tense*
 - Chapter 4: The Countryside Forest — *LO4: Word Forms*
 - Chapter 5: The Jungle's Edge — *LO5: Full Application*
- **Animated Lesson Videos** for each chapter introducing the grammar concept
- **Vocabulary Cards** with phonetic spelling, meaning, and story context
- **Word of the Day** feature with pronunciation playback and example use
- **Chapter Progress Tracking** with persistent journey state

Speaking & Assessment System

- **On-Device Whisper Transcription** via whisper.rn — no audio leaves the device
- **Multi-Question Types:** Identify, Build (fill-in-the-blank), Speak (read-aloud), Choice (read-only options), and Rival (correct a wrong sentence)
- **Three-Factor Scoring:**
 - Content (50%) — does the answer match the expected response
 - Pronunciation (40%) — Whisper confidence mapped through a child-speech-tuned curve
 - Fluency (10%) — pacing and naturalness

- **Pass Threshold of 80** with tiered feedback (Amazing!, Great job!, Almost there!, Keep trying!, Let's practice!)
- **Phoneme-Level Feedback** using the bundled CMU dictionary and Needleman–Wunsch alignment to surface the exact sounds a learner missed
- **Adaptive Hint System** — three escalating hints, with the answer revealed after the maximum wrong attempts
- **15-Second Auto-Stop Recording** with silence detection

Gamification & Rewards

- **Interactive Story Map** showing chapter unlock progression across the city
- **Animal Companion Rewards** — each chapter reunites Alex with a new animal friend (Skunk, Squirrel, Rooster, Owl, Parrot)
- **Badge & Achievement System** powered by lib/badgeEngine.ts for milestone tracking
- **Final Quest** unlocked after all five chapters are completed
- **Certificate of Completion** awarded at journey's end
- **Reunion Sequence** celebrating every animal Alex has helped
- **Animated Alex Character** with mood-driven Lottie reactions and speech bubbles

Audio & Media

- **Embedded Lesson Videos** with custom video player and orientation handling
- **Background Music Player** with chapter-specific tracks
- **Alex Speech Playback** for in-character narration via expo-speech
- **Vocabulary Audio Pronunciation** with replayable word samples
- **expo-audio / expo-av Recording** with automatic resource cleanup

Teacher Mode

- **PIN-Gated Teacher Portal** at /teacher with secure access
- **Assessment History Dashboard** showing every attempt, score breakdown, and transcript
- **Rubric Reference Data** (lib/teacherRubricData.ts) for grading guidance
- **Per-Student Analytics** including problem sounds and common errors

Onboarding

- **Animated Intro Cinematic** introducing Alex's story
- **Step-by-Step Tutorial** teaching learners how to record, listen, and answer
- **Quest Prologue Video** before the final challenge

UI/UX Design

Design Principles

- **Child-Friendly Interface:** Bright, encouraging color palettes per chapter (amber bakery, green forest, jungle teal, etc.)
- **Accessibility First:** Large tap targets, readable typography, clear feedback states
- **Responsive Layout:** Adapts to phone and tablet screen sizes
- **Smooth Animations:** Moti and Reanimated micro-interactions for every transition

Theme System

- **NativeWind (Tailwind) Theming:** Consistent design tokens across the app
- **Per-Chapter Accent Colors:** Each location has its own gradient and palette
- **Custom Typography:** Optimized for young readers
- **Lottie Animations:** Alex's emotional reactions and celebration moments

Security & Privacy

Data Protection

- **On-Device Speech Processing:** Whisper transcription runs locally — student audio is never uploaded
- **Local Progress Storage:** Zustand stores ([store/gameStore.ts](#), [store/teacherStore.ts](#)) persist progress on the device
- **PIN-Gated Teacher Access:** Sensitive assessment history is locked behind a teacher PIN
- **Permission Transparency:** Microphone and speech-recognition prompts declared in [app.json](#)

Privacy Features

- **Minimal Data Collection:** No accounts, no telemetry, no cloud sync of student speech

- **Permission-Scoped Audio:** Recording only active during answer capture
- **Auto-Cleanup:** Audio buffers released after every assessment cycle

Performance Optimizations

Efficiency Features

- **Native Whisper Module:** Compiled whisper.rn runs faster than JS-based recognizers
- **Fallback Recognizer:** expo-speech-recognition available when Whisper init fails
- **Lazy Asset Loading:** Chapter videos and audio loaded only when entered
- **State Preservation:** Zustand survives reloads; progress resumes mid-chapter

Resource Management

- **Audio Resource Cleanup:** Recorders fully released between questions to prevent memory leaks
- **Animation Lifecycle Handling:** Lottie and Reanimated values properly disposed
- **Patch-Package Stability:** Custom patches applied via postinstall for native module fixes

Supported Devices

Minimum Requirements

- **Android Version:** 7.0 (API level 24)
- **iOS Version:** 14.0+
- **RAM:** 3GB minimum, 4GB recommended (Whisper model)
- **Storage:** ~250MB for app, model, and audio assets
- **Microphone:** Required for all speaking activities

Optimal Experience

- **Android Version:** 11.0+ (API level 30+)
- **iOS Version:** 16.0+
- **RAM:** 4GB or more
- **Display:** 5.5" or larger screen
- **Audio:** Built-in speakers or headphones for lesson playback

Note: Alex's Quest requires a development build — the app uses native modules (whisper.rn, expo-speech-recognition, expo-audio). Expo Go is not supported.

Acknowledgments

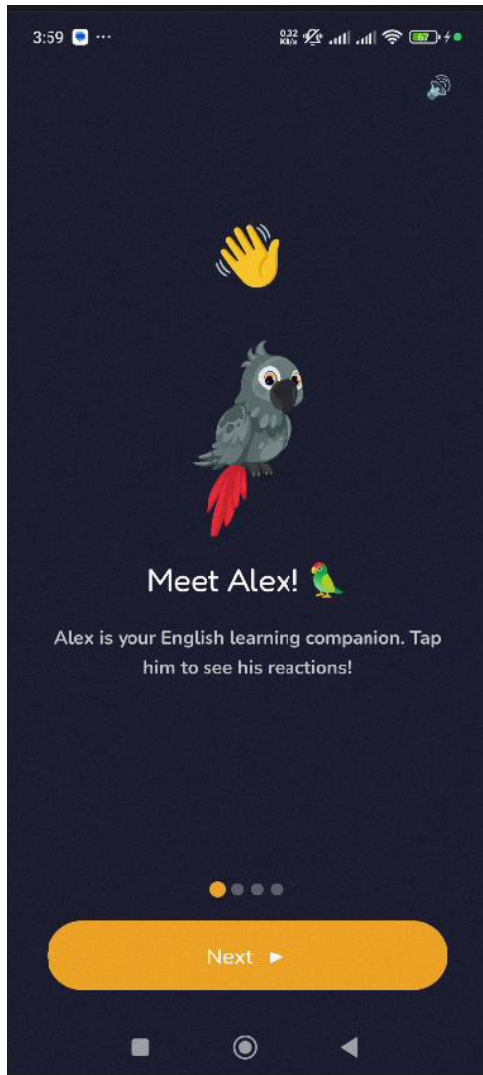
Technologies Used

- **Expo SDK 55 / React Native 0.83 / React 19** — Cross-platform runtime
- **expo-router** — File-based, typed navigation
- **NativeWind (Tailwind)** — Utility-first styling
- **Zustand** — Lightweight global state
- **whisper.rn** — On-device Whisper transcription with confidence scoring
- **expo-speech-recognition** — Fallback recognizer
- **expo-audio / expo-av** — Recording and playback
- **Moti + React Native Reanimated** — Declarative animations
- **Lottie React Native** — Vector character animations
- **CMU Pronouncing Dictionary** — Phoneme reference data
- **EAS Build** — Cloud build and release pipeline

Educational Content

- Five-chapter story arc and learning objectives crafted around English-grammar curriculum milestones
- Vocabulary, hints, and rival lines authored to scaffold attempts and reward correct construction
- Phoneme-level feedback curated for common Filipino-English child-speech challenges

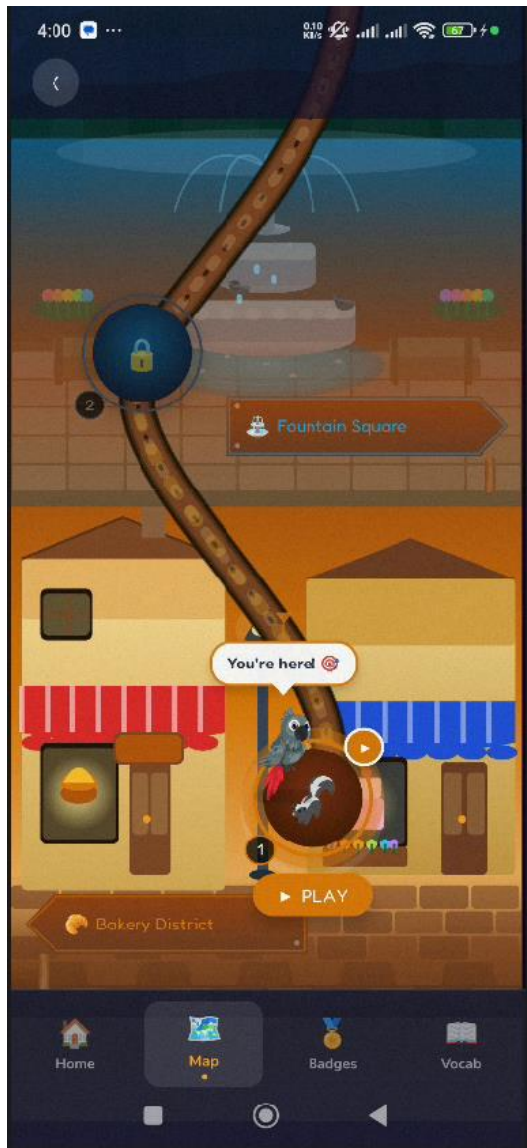
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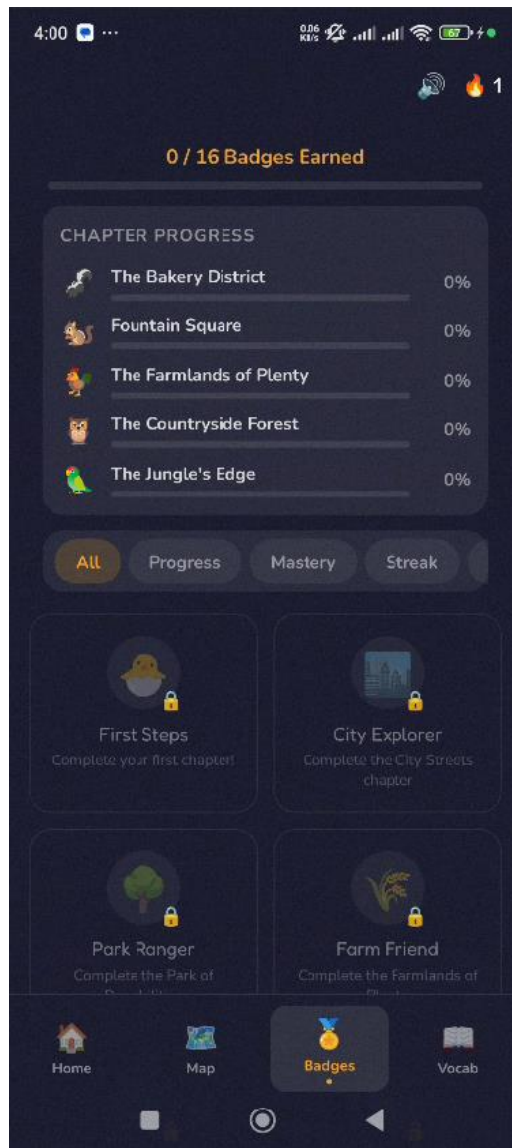
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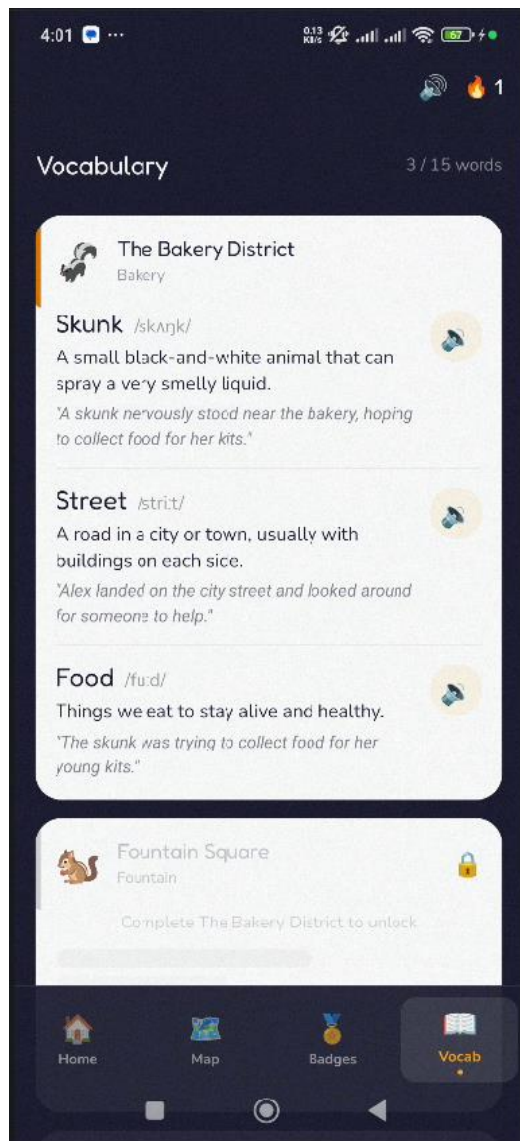
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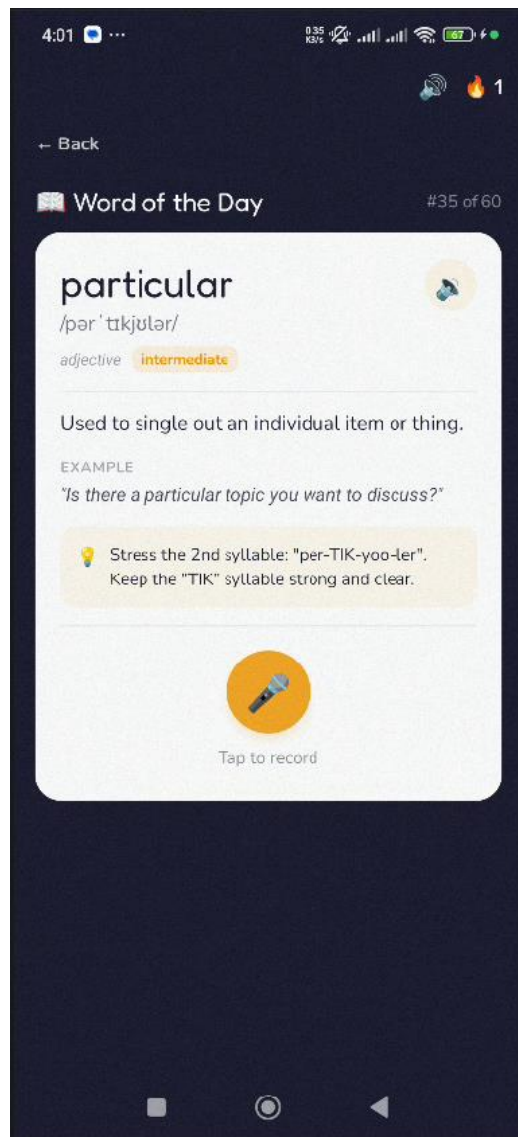
MAP



BADGES



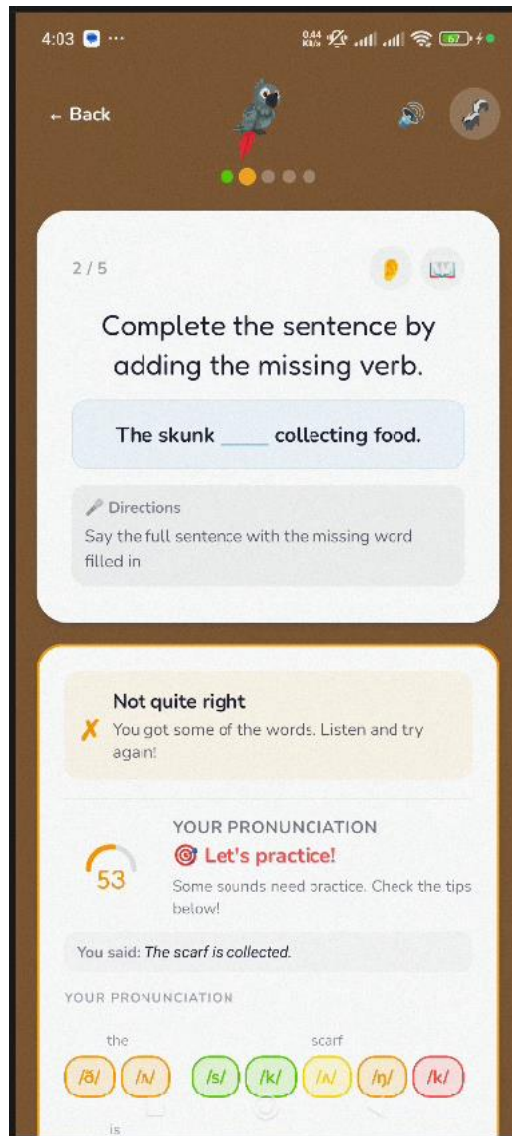
VOCAB



WORD OF THE DAY



CHALLENGES



ASSESSMENT